Unit: I - Differential Calculus

- Diff^{n} of Hypesbolic & Invesse Hypesbolic Function.
- Successive Differentiation.
- Standard foams
- Leibnitz's theosem & applications
- Powes Sesies
- Explanation Expansion of functions.
- Taylos's & Maclausin's sesies.
- Cusvatuse
- Radius of cusvatuse fos Castesian Cusve with application.

⇒ Hypesbolic Function:

- let x be a Real (OR) complex quautity, then the hypesbolic "sine" of x is denoted by "sinhx" and the hypesbolic "cosine" of x is denoted by "coshx" and the hypesbolic "tangent" of x is denoted by "tanhx" are defined by,

$$\cosh x = \frac{e^{x}+e^{-x}}{2}, \quad \text{Pesiod} : 2n\pi i$$

$$\sinh x = \frac{e^{x}-e^{-x}}{2}, \quad \text{Pesiod} = 2n\pi i^{0}$$

$$\tanh x = \frac{e^{x}-e^{-x}}{e^{x}+e^{-x}}, \quad \text{Pesiod} : n\pi i^{0}$$

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Circle - $x^2+y^2=1$

$x = r\cos\theta$, $y = r\sin\theta$

Any point on the circle can be expressed in the form of $\sin\theta, \cos\theta$

- Any point on the hyperbola can be represented by $\sinh x$ a $\cosh x$.

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- The reciprocals of these functions are cosechx, sechx and cothx respectively and are defined by,

$$\operatorname{cosech} x = \frac{2}{e^x - e^{-x}}, \quad x \in R - \{0\}$$

$$\operatorname{sech} x = \frac{2}{e^x + e^{-x}}, \quad x \in R - \{0\}$$

$$\coth x = \frac{e^x + e^{-x}}{e^x - e^{-x}}, \quad x \in R - \{0\}$$

⇒ Direct circular Function and hyperb Function.

- circular function is define for all Real no., not only x, here it's a complex quantity but we can mak for real quantity as well.

⇒ <u>circular Function:</u>

- If we can define "sine" and "cosin for any complex Variable then it wil become circular function.

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⇒ Direct Circular Function:

- If we can define sine & cosine in exponential term then it's called direct circular function.

eg. $z = x + iy$

$$\sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots$$

$$\cos z = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \cdots$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$e^{iz} = 1 + iz + \frac{(iz)^2}{2!} + \cdots$$

because $e^{iz} = \cos z + i \sin z$ — (1)

$$e^{-iz} = \cos z - i \sin z \quad — (2)$$

(1) + (2), $e^{iz} + e^{-iz} = 2\cos z$

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}$$

(1) − (2), $e^{iz} - e^{-iz} = 2i \sin z$

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}$$

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$\sin(z + 2n\pi) = \sin z$

$\sin z \cdot \cos 2n\pi + \cos z \cdot \sin 2n\pi$

$1 \qquad\qquad 0$

$\cos(z + 2n\pi) = \cos z$

$\tan(z + n\pi) = \tan z$

So here we know that

$$\cos^2 x + \sin^2 x = 1$$
$$\Rightarrow \cosh^2 x - \sinh^2 x = 1$$

$$\cosh x = \frac{e^x + e^{-x}}{2} \quad \& \quad \sinh x = \frac{e^x - e^{-x}}{2}$$

⇒ Identities:

1) $\cosh^2 x - \sinh^2 x = 1$
2) $\cosh^2 x + \sinh^2 x = \cosh 2x$
3) $1 - \tanh^2 x = \operatorname{sech}^2 x$
4) $\coth^2 x - 1 = \operatorname{cosech}^2 x$
5) $\sinh(x \pm y) = \sinh x \cdot \cosh y \pm \cosh x \cdot \sinh y$
6) $\cosh(x \pm y) = \cosh x \cdot \cosh y \pm \sinh x \cdot \sinh y$

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eg: If $\sinh x = u$

$\Rightarrow \cosh x = \sqrt{1+u^2}$

If $\cosh x = u$

$\Rightarrow \sinh x = \sqrt{u^2-1}$

If $\tanh x = u$

$\sinh x = \pm \frac{u}{\sqrt{1-u^2}}$, $\cosh x = \frac{1}{\sqrt{u^2-1}}$ $\frac{1}{\sqrt{1-u^2}}$

$\coth x = \frac{1}{u}$, $\operatorname{sech} x = \sqrt{1-u^2}$

$\operatorname{cosech} x = \pm \frac{\sqrt{1-u^2}}{u}$

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$\tanh x = u$

$1 - \tanh^2 x = \operatorname{sech}^2 x$

$1 - u^2 = \operatorname{sech}^2 x$

$\sqrt{1-u^2} = \operatorname{sech} x$

$\cos h x = \dfrac{1}{\sqrt{1-u^2}}$

$\rightarrow \cos h^2 x - \sin h^2 x = 1$

$\left(\dfrac{1}{\sqrt{1-u^2}}\right)^2 - \sin h^2 x = 1$

$\sin h^2 x = \dfrac{1}{1-u^2} - 1$

$= \dfrac{1 - 1 + u^2}{1-u^2}$

$\sin h^2 x = \dfrac{u^2}{1-u^2}$

$\sin h x = \pm \dfrac{u}{\sqrt{1-u^2}}$

$\cosh^2 x - \sinh^2 x = 1$

$\cosh^2 x - 1 = \sinh^2 x$

$\left(\dfrac{1}{\sqrt{1-u^2}}\right)^2 - 1 = \sinh^2 x$

$\dfrac{1}{1-u^2} - 1 = \quad ''$

$\dfrac{1 - 1 + u^2}{1-u^2} \quad ''$

$\sinh^2 x = \dfrac{u^2}{1-u^2}$

$\sinh x = \dfrac{\pm u}{\sqrt{1-u^2}}$

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Relation between circulas function and Hyperbolic functions.

$$\boxed{\cos(ix) = \cosh x}$$

$$\cos iz = \frac{e^{iz} + e^{-iz}}{2}$$

Put $z = ix$

$$\cos(ix) = \frac{e^{i(ix)} + e^{-i(ix)}}{2}$$

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$$\cos(ix) = \frac{e^{i^2 x} + e^{-i^2 x}}{2}$$

$$\cos(ix) = \frac{e^{-x} + e^{+x}}{2}$$

$$\boxed{\cos(ix) = \cosh x}$$

2) $\sin(ix) = i\sinh x$

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}$$

Put $z = ix$

$$\sin(ix) = \frac{e^{i(ix)} - e^{-i(ix)}}{2i}$$

$$= \frac{e^{i^2 x} - e^{-i^2 x}}{2i}$$

$$= \frac{e^{-x} - e^{+x}}{2i}$$

$$= -\frac{(e^{x} - e^{-x})}{2} \times \frac{1}{i} \times \frac{i}{i}$$

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$$\sin(ix) = i \times \left[\left(\frac{e^{x} - e^{-x}}{2}\right)\right] \qquad (\because i^2 = -1)$$

$$\boxed{\sin(ix) = i \sinh x}$$

3) $\tan(ix) = i \tanh x$

$$\tan(ix) = \frac{\sin ix}{\cos ix} \qquad \text{so,}$$

$$= \frac{i \sinh x}{\cosh x}$$

$$\boxed{\tan(ix) = i \tanh x}$$

4) $\sec(ix) = \operatorname{sech} x$

$$\sec(ix) = \frac{1}{\cos(ix)}$$

$$= \frac{1}{\cosh x}$$

$$\sec = \operatorname{sech} x$$

so that,

$$\boxed{\sec(ix) = \operatorname{sech} x}$$

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5) $\operatorname{cosec}(ix) = -i\,\operatorname{cosech} x$

$$\operatorname{cosec}(ix) = \frac{1}{\sin(ix)}$$

$$= \frac{1}{i\sinh x} \times \frac{i}{i}$$

$$= \frac{i}{-\sinh x}$$

$$\boxed{\operatorname{cosec}(ix) = -i\,\operatorname{cosech} x}$$

6) $\cot(ix) = -i\coth x$

$$\cot(ix) = \frac{\cos ix}{\sin ix}$$

$$= \frac{\cosh x}{i\sinh x} \times \frac{i}{i}$$

$$= \frac{i\cosh x}{-\sinh x}$$

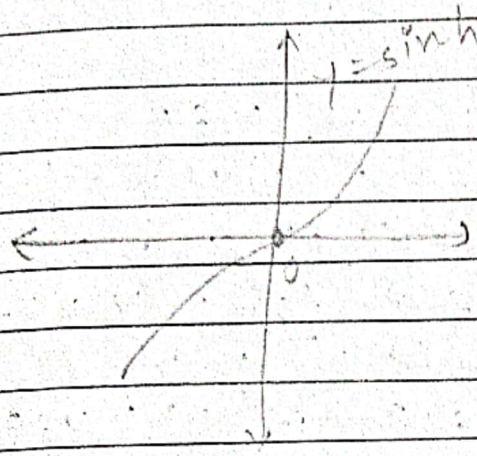
$$= -i\coth x$$

$$\boxed{\cot(ix) = -i\coth x}$$

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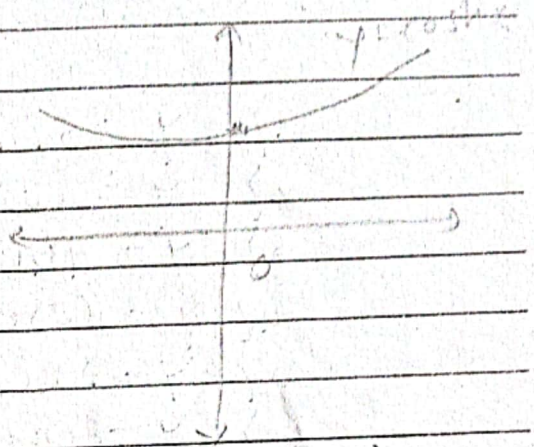
⇒ Range and Domain of Hyperbolic Functions with their graphs.

1) $y = \sinh x$



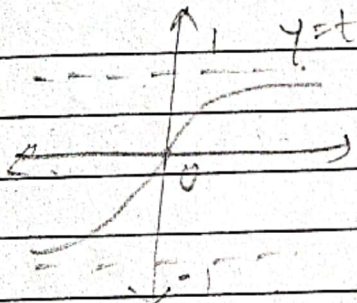
Range: $(-\infty, \infty)$
Domain: $(-\infty, \infty)$

2) $y = \cosh x$



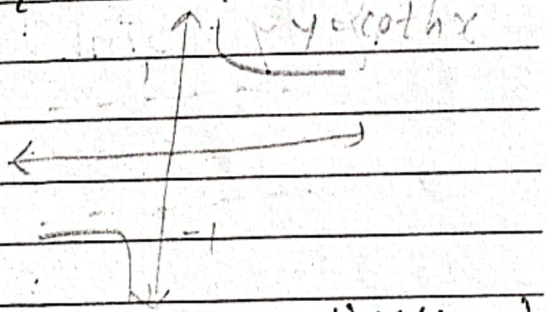
Range: $[1, \infty)$
Domain: $(-\infty, \infty)$

3) $y = \tanh x$



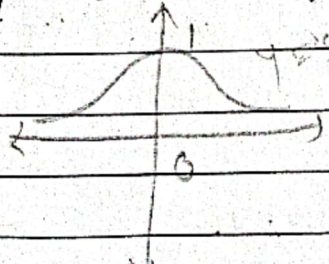
Range: $(-1, 1)$
Domain: $(-\infty, \infty)$

4) $y = \coth x$



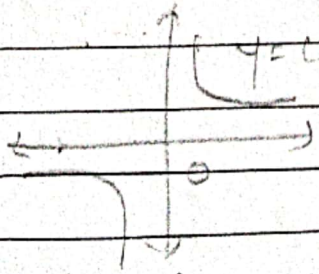
Range: $(-\infty, -1) \cup (1, \infty)$
Domain: $(-\infty, \infty) - (0)$

5) $y = \operatorname{sech} x$



Range: $(0, 1)$
Domain: $(-\infty, \infty)$

6) $y = \operatorname{cosech} x$



Range: $(-\infty, \infty) - [0]$
Domain: $(-\infty, \infty) - [0]$

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⇒ Inverse Hyperbolic Function:

- The inverse hyperbolic of sine is denoted by $\sinh^{-1}x$ and other inverse hyperbolic functions are defined by,

1) $y = \sinh^{-1}x$; $x \in R$, $y \in R$
2) $y = \cosh^{-1}x$; $x \in [1, \infty)$; $y \in [0, \infty)$
3) $y = \tanh^{-1}x$; $x \in (-1, 1)$; $y \in R$
4) $y = \operatorname{cosech}^{-1}x$; $x \in R - \{0\}$; $y \in R - \{0\}$
5) $y = \operatorname{sech}^{-1}x$; $x \in (0, 1]$; $y \in (0, \infty)$
6) $y = \coth^{-1}x$; $x \in R - [-1, 1]$; $y \in R - \{0\}$

⇒ Inverse hyperbolic functions in terms of logarithmic Functions.

1) $y = \sinh^{-1}x$

let $y = \sinh^{-1}x \Rightarrow x = \sinh y$

$$x = \frac{e^y - e^{-y}}{2}$$

$$2x = e^y - e^{-y}$$

both side Multiply by e^y.

$$2xe^y = e^y \cdot e^y - e^{-y} e^y$$

$$2xe^y = e^{y+y} - e^{-y+y}$$

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$$2xe^y = e^{2y} - 1$$

$$e^{2y} - 2xe^y - 1 = 0$$

$$(e^y)^2 - 2xe^y - 1 = 0$$

comparing it with $ax^2 + bx + c = 0$.

$$a = 1, \quad b = -2x, \quad c = -1$$

$$e^y = \frac{-(-2x) \pm \sqrt{(-2x)^2 - 4(1)(-1)}}{2(1)}$$

$$e^y = \frac{2x \pm \sqrt{4x^2 + 4}}{2}$$

$$e^y = x \pm \sqrt{x^2 + 1}$$

$$y = \log(x \pm \sqrt{x^2 + 1})$$

$$\boxed{\sinh^{-1} x = \log(x \pm \sqrt{x^2 + 1})}$$

24 $y = \cosh^{-1} x$

By taking Positive radicals in $\sinh^{-1} x$,

$$y = \sinh^{-1} x = \log(x + \sqrt{x^2 + 1})$$

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$\cosh^{-1} x = \ln\left(x + \sqrt{x^2 - 1}\right), \quad x \geq 1$

($\cosh^{-1} x > 0$, is principal value to avoid multi value we restrict our selves.)

3) $y = \tanh^{-1} x$

$\tanh^{-1} x = \frac{1}{2} \log\left(\frac{1+x}{1-x}\right), \quad -1 < x <$

4) $\coth^{-1} x = y$

$\coth^{-1} x = \frac{1}{2} \log\left(\frac{x+1}{x-1}\right), \quad \begin{matrix} x > 1, \\ x < -1 \end{matrix}$

5) $y = \operatorname{sech}^{-1} x$

$\operatorname{sech}^{-1} x = \log\left(\frac{1}{x} + \sqrt{\frac{1}{x^2} - 1}\right), \quad 0 < x$

$\operatorname{sech}^{-1} x > 0$ is Principal value

6) $y = \operatorname{cosech}^{-1} x$

$\operatorname{cosech}^{-1} x = \ln\left(\frac{1}{x} + \sqrt{\frac{1}{x^2} + 1}\right), \quad x \neq$

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Proof of $y = \cosh^{-1} x$

Let $x = \cosh y$

$$= \frac{e^y + e^{-y}}{2}$$

Now Multiplying & divide by e^y

$$x = \frac{e^y + e^{-y}}{2} \times \frac{e^y}{e^y}$$

$$x = \frac{e^{2y} + 1}{2e^y}$$

$$2e^y x = e^{2y} + 1$$

$$e^{2y} - 2e^y x + 1 = 0$$

$$(e^y)^2 - 2xe^y + 1 = 0$$

$$e^y = \frac{2x + \sqrt{4x^2 - 4}}{2}$$

$$e^y = x + \sqrt{x^2 - 1}$$

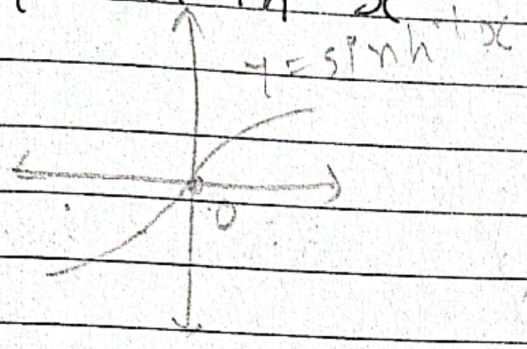
$$y = \log (x + \sqrt{x^2 - 1})$$

$$\cosh^{-1} x = \log (x + \sqrt{x^2 - 1})$$

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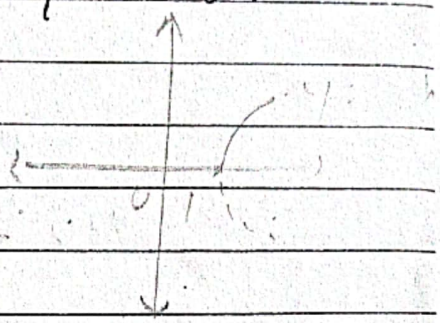
⇒ Range and Domain of Inverse Hyperbolic Function with their Graphs.

1) $y = \sinh^{-1} x$



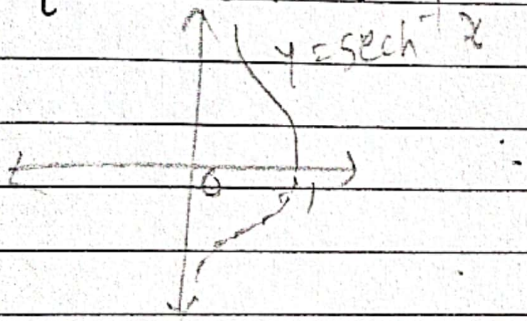
Range: $(-\infty, \infty)$
Domain: $(-\infty, \infty)$

2) $y = \cosh^{-1} x$



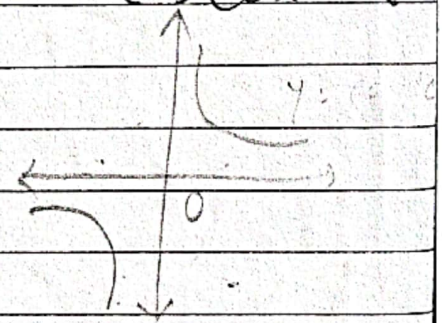
Range: $[0, \infty)$
Domain: $[1, \infty)$

3) $y = \operatorname{sech}^{-1} x$



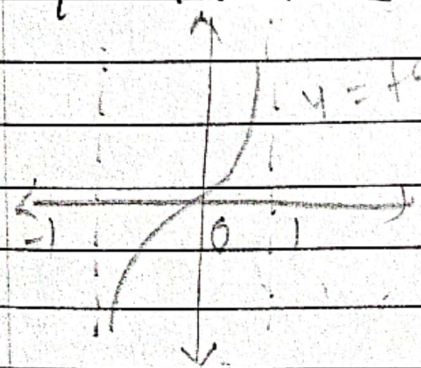
Range: $[0, \infty)$
Domain: $(0, 1]$

4) $y = \operatorname{cosech}^{-1} x$



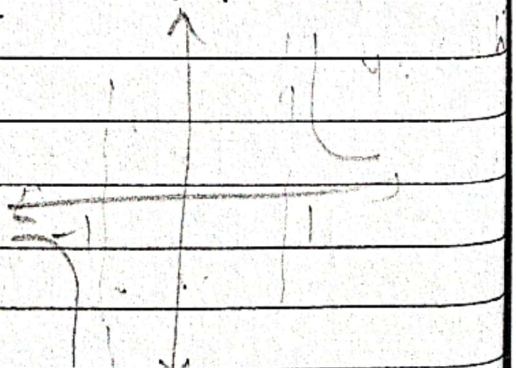
Range: $(-\infty, \infty) - [0$
Domain: $-\infty < x < \infty$
$-\{0\}$

5) $y = \tanh^{-1} x$



Domain ~~Range~~: $(-1, 1)$
Range ~~Domain~~: $(-\infty, \infty)$

6) $y = \coth^{-1} x$



Range: $(-\infty, \infty) - (0$
Domain: $(-\infty, -1) \cup (1, \infty$

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